Experiment-1

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FRAME 1: INSTRUCTION PAGE

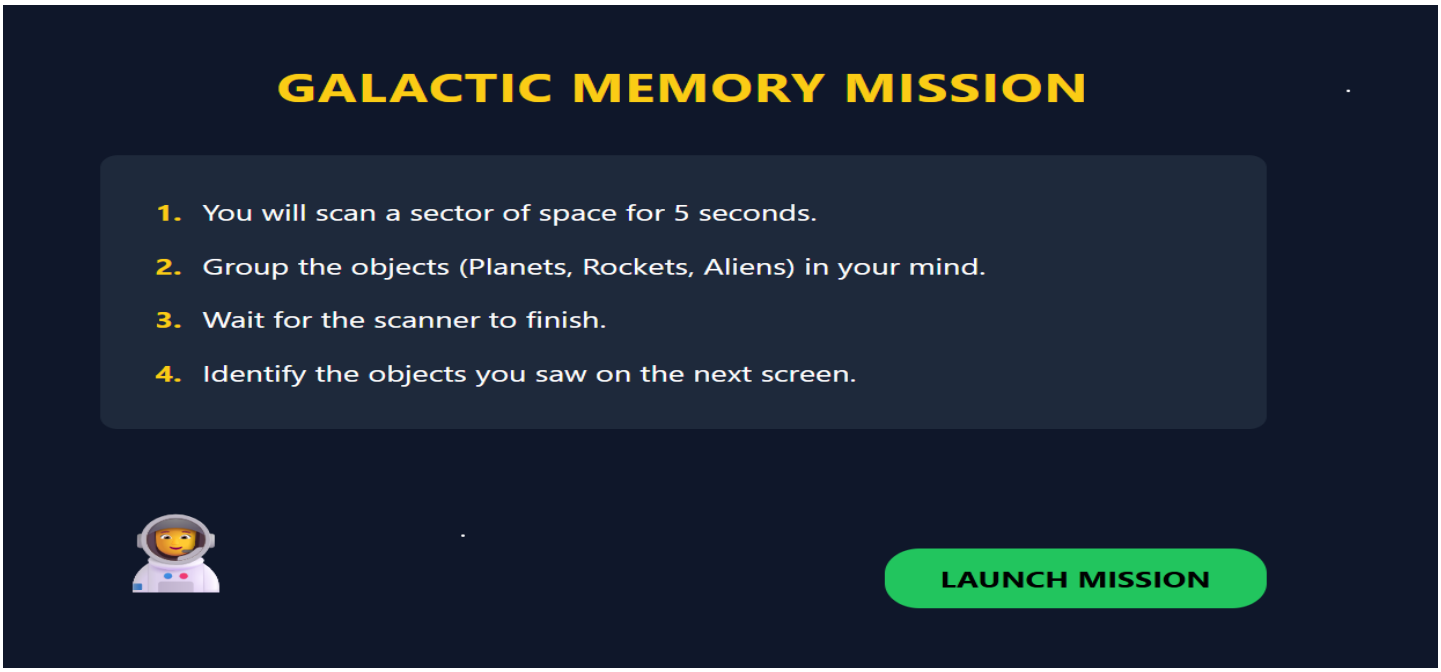
Screen Title: GALACTIC MEMORY MISSION

UI Description:

- **Background:** Deep blue space theme with faint twinkling stars.
- **Central Title:** "GALACTIC MEMORY MISSION" in large, yellow futuristic font.
- **Mascot:** A cartoon Astronaut floating on the bottom left.
- **Button:** A large green "LAUNCH" button at the bottom.
- **Text Body:**
    1. You will scan a sector of space for **5 seconds**.
    2. **Group the objects** (Planets, Rockets, Aliens) in your mind.
    3. Wait for the transmission to end.
    4. **Identify** the objects you saw on the next screen.
    5. *Tip: Remember the colors!*

Chunking Analysis of the Instruction Page:

- **Cognitive Preparation:** The instructions explicitly hint at a strategy ("Group the objects"), priming the user to use chunking before the task begins.
- **Step-by-Step Flow:** Information is broken into 5 short, numbered lines. This prevents "Instructional Blindness" where users skip large blocks of text.
- **Visual Hierarchy:** The "LAUNCH" button is distinct and isolated, drawing the eye naturally to the bottom after reading the text.



FRAME 2: CHUNKING PHASE (The Stimulus)

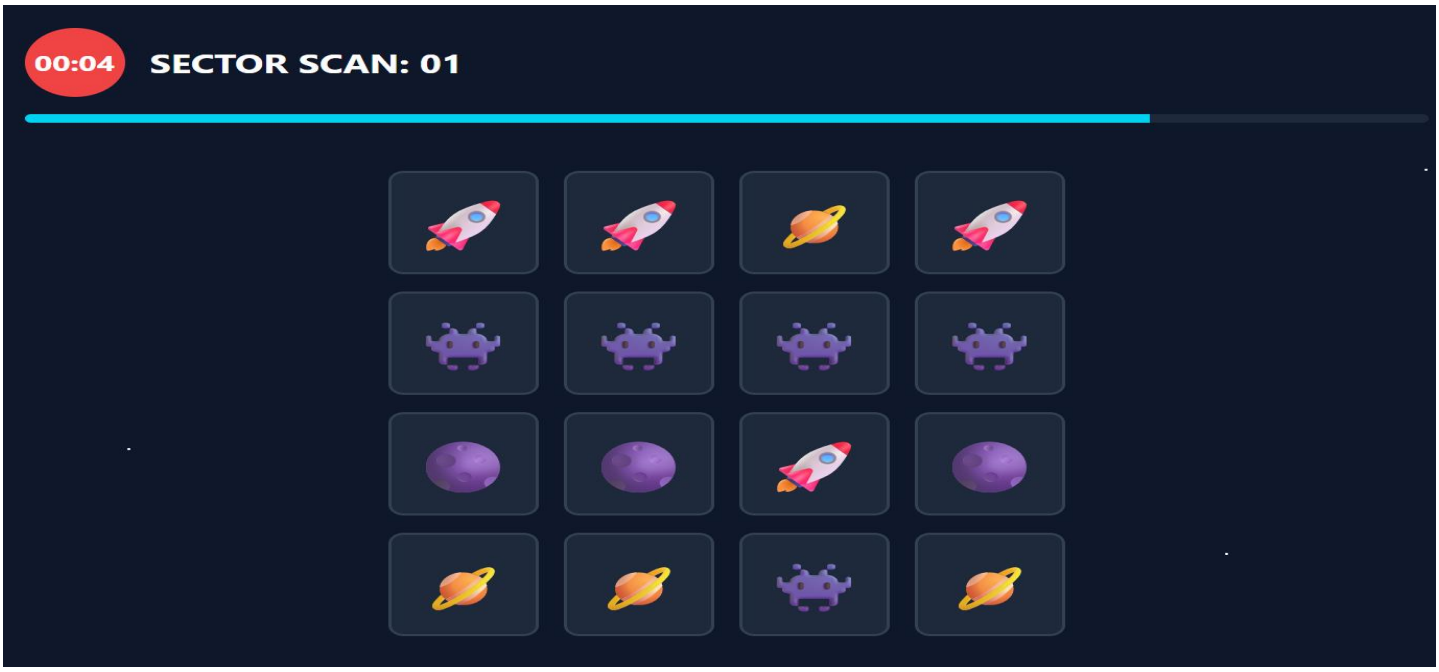
Screen Title: SECTOR SCAN: 01

UI Description:

- **Timer (Top Left):** A red digital countdown clock showing **00:05**.
- **Progress Bar:** A cyan bar shrinking rapidly from right to left.
- **The Grid (Center):** A 5x4 grid of icons on dark tiles.
    - **Row 1 (Red Items):** Red Planet, Red Rocket, Red Alien, Red Flag.
    - **Row 2 (Mixed):** Blue Rocket, Yellow Star, Red Planet, Blue Rocket.
    - **Row 3 (Cluster):** Three "Green Alien" icons grouped together.
- **Mascot:** The Astronaut holds a radar screen.

Analysis of the Chunking Phase Screen:

- **Categorical Chunking:** The icons are not random; they are designed to be "chunked" by color (all Red items) or by type (all Aliens). This tests if users recall the *category* (e.g., "I saw lots of red things") better than individual items.
- **Urgency Constraints:** The "00:05" timer and shrinking bar impose a "cognitive bottleneck," forcing the brain to abandon detailed scanning and switch to pattern recognition (chunking).
- **Grid Layout:** The 5x4 structure provides a clean mental map, allowing users to remember spatial locations (e.g., "The aliens were in the bottom left").



FRAME 3: RECALL PHASE (Selection)

Screen Title: REPORT YOUR FINDINGS

UI Description:

- **Instruction Text:** "Select the objects present in the Sector."
- **The Selection Grid:** A set of 8 large distinct icons.
  - *Correct Options:* Red Planet, Green Alien, Blue Rocket.
  - *Distractors (False Options):* Purple Comet, Orange Sun, Pink UFO (These were *not* in the previous screen).
- **Interaction:** Circular "Radio Buttons" under each icon.
- **Action Button:** "TRANSMIT DATA" (Submit).

Analysis of the Selection Phase:

- **Recognition vs. Recall:** This screen tests recognition memory. By providing distractors (items that fit the theme but weren't shown), it forces the user to verify their memory chunks.
- **The "Lure" Effect:** The distractors are designed to be visually similar (e.g., a Pink UFO looks like the Green Alien). If a user chunked only by shape ("I saw an alien") but forgot the color, they might choose the wrong one.
- **Clean Affordance:** The radio buttons clearly indicate that a selection is required, reducing friction in the interaction.

FRAME 4: RESULT PAGE

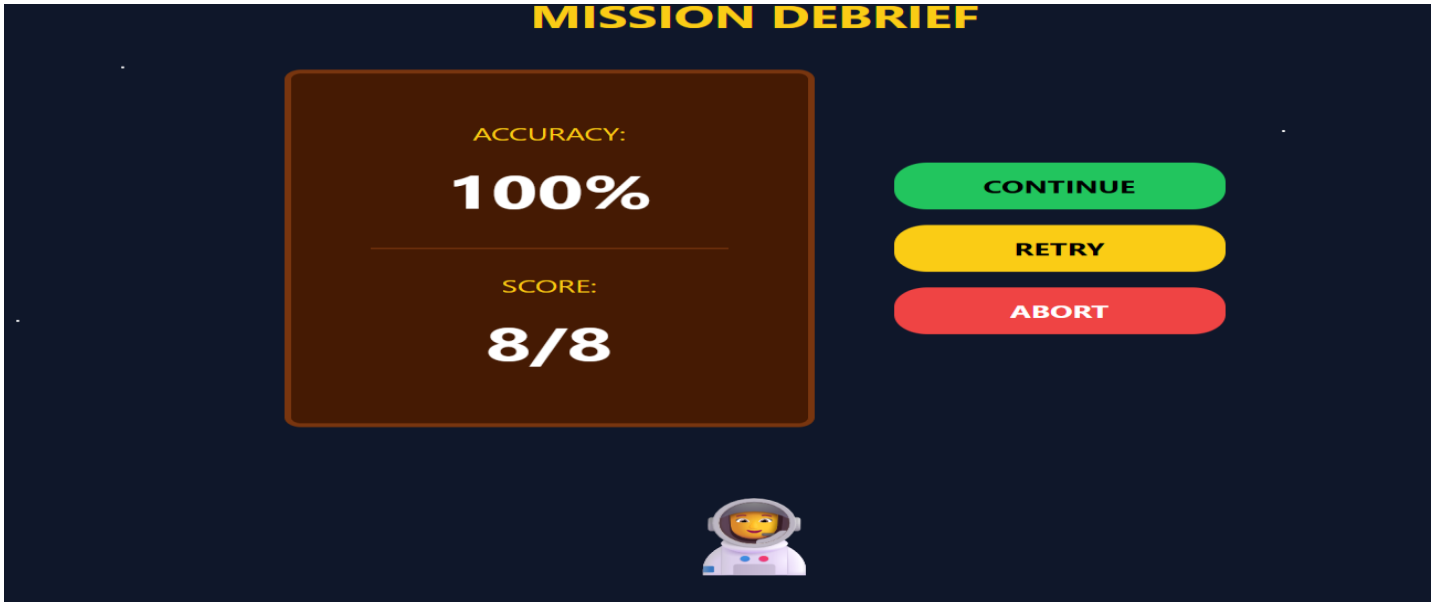
Screen Title: MISSION DEBRIEF

UI Description:

- **Main Visual:**
  - *If High Score:* The Astronaut planting a flag on a moon.
  - *If Low Score:* The Astronaut floating away with a confused face.
- **Score Board:** A holographic panel displaying:
  - **ACCURACY:** 80%
  - **ITEMS FOUND:** 4/5
- **Feedback Text:** "Excellent Pattern Recognition, Cadet!"
- **Action Buttons:**
  - [NEXT SECTOR] (Yellow) - Leads to a harder level.
  - [RETRY MISSION] (Gray) - Restarts the current level.
  - [ABORT] (Red) - Exit.

Analysis of the Result Screen:

- **Immediate Feedback Loop:** Users instantly see the connection between their strategy (chunking) and the outcome (score).
- **Gamification:** The title "Mission Debrief" and the visual of the astronaut maintain the narrative, transforming a boring memory test into an engaging challenge.
- **User Control:** Providing "Retry" allows users to practice their chunking strategy immediately, reinforcing the learning curve.



Prototype Link:

https://www.figma.com/make/tK3cPijSwI3U7VuYQw4Wo5/Galactic-Memory-Mission-Design?fullscreen=1&t=DLhue7MqfiiI8C9o-1